

Carnegie Mellon University
MSCF Program

Matrices & Linear Algebra – Part 1 (MSCF Math Prep)

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What is Linear Algebra?

Systems of linear equations play an important role in computational finance (and in many other disciplines as well).

Matrices allow us to write a system of linear equations in a very compact form.

It is useful to think about matrices as linear mappings (or *linear transformations*) between spaces of vectors.

Having a more compact notation and a more general and abstract framework can lead to increased conceptual understanding and insight, and also to increased computational efficiency!

The study of *Linear Transformations* is what *Linear Algebra* is all about.

Where will We Use Matrices and Linear Algebra in MSCF?

Here are just a few places where matrices and linear algebra will appear in the MSCF Program:

- ▶ In Portfolio Optimization and Asset Allocation.
- ▶ Finding discount factors from a list of bond prices.
- ▶ Computing the variance of a linear combination of random variables in terms of the variances and covariances of the individual random variables.
- ▶ Taking a list of correlated random variables and producing a list of uncorrelated random variables that “conveys the same information” as the original list.
- ▶ The first and second Fundamental Theorems of Asset Pricing.
- ▶ Regression and related statistical and machine learning concepts.
- ▶ Principal Components

Matrices

Matrix: A *matrix* (plural *matrices*) is a rectangular array of real (or complex) numbers – called *entries* – arranged in rows and columns.

We shall make the convention that, unless stated otherwise, the entries are understood to be real numbers. There will be some situations where it is convenient (or even essential) to allow the entries to be complex numbers – in such cases, it will always be stated explicitly that we are allowing complex entries.

A matrix with m rows and n columns is said to be an $m \times n$ matrix. The set of all $m \times n$ matrices will be denoted by

$$M^{m \times n}.$$

Matrices (Continued)

For $A \in M^{m \times n}$ we use the notation A_{ij} to indicate the entry in row i and column j .

The entries of a matrix are generally enclosed by parentheses or square brackets. We shall use parentheses.

Example:
$$A = \begin{pmatrix} 2 & -1 & 3 \\ 5 & 0 & \sqrt{2} \end{pmatrix}$$

Observe that $A \in M^{2 \times 3}$ and that, for example, $A_{23} = \sqrt{2}$, $A_{12} = -1$.

Column Vectors

A matrix with only one column, e.g. $\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$ is called a *column vector*.

We write $V^n = M^{n \times 1}$ for the set of column vectors having n entries.

Since a column vector has only one column, it is customary to use only one index on each entry (to designate the row), i.e. for $x \in V^n$ we write

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{pmatrix}.$$

Scalars

In the context of matrices and vectors, we use the term *scalar* to indicate a real (or complex) number.

If we are allowing only real entries in our matrices, then scalar means a *real number*.

If we are allowing complex entries in our matrices, then scalar means a *complex number*.

Equality of Matrices, The Zero Matrix

Let A and B be matrices. When we write $A = B$ we mean that A and B have the same size, say $m \times n$, and that

$$A_{ij} = B_{ij} \text{ for all } i = 1, 2, \dots, m, j = 1, 2, \dots, n.$$

It is convenient to have a special symbol to indicate the $m \times n$ matrix all of whose entries are zero; we denote this matrix by $0_{m \times n}$. For example

$$0_{2 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

When the size should be clear from context, we write 0 in place of $0_{m \times n}$.

Matrix Addition

We can add matrices of the same size in a straightforward fashion. Let $A, B \in M^{m \times n}$ be given. We define $A + B \in M^{m \times n}$ by

$$(A + B)_{ij} = A_{ij} + B_{ij} \quad \text{for all } i = 1, 2, \dots, m, j = 1, 2, \dots, n.$$

Example:
$$\begin{pmatrix} 2 & 1 \\ 3 & 4 \\ -1 & 6 \end{pmatrix} + \begin{pmatrix} 0 & -2 \\ 1 & 6 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 4 & 10 \\ 2 & 11 \end{pmatrix}$$

Since addition of matrices is defined entry-wise, the familiar properties of addition are easily seen to hold.

Multiplication of a Matrix by a Scalar

Let α be a scalar and $A \in M^{m \times n}$ be given. We define $\alpha A \in M^{m \times n}$ by

$$(\alpha A)_{ij} = \alpha A_{ij} \text{ for all } i = 1, 2, \dots, m, j = 1, 2, \dots, n.$$

Example: $3 \begin{pmatrix} 1 & 2 \\ 4 & -3 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 12 & -9 \end{pmatrix}$

The basic properties of this operation (and interaction with matrix addition) follow easily from the definitions and the properties of ordinary addition and multiplication.

Transpose of a Matrix

Let $A \in M^{m \times n}$ be given. The *transpose* of A , denoted by $A^\top$ is the $n \times m$ matrix defined by

$$(A^\top)_{ij} = A_{ji} \quad \text{for all } i = 1, 2, \dots, n, j = 1, 2, \dots, m.$$

Example:

$$\begin{pmatrix} 2 & 3 & -1 \\ 0 & 5 & 6 \end{pmatrix}^\top = \begin{pmatrix} 2 & 0 \\ 3 & 5 \\ -1 & 6 \end{pmatrix}$$

Basic Properties of the Transpose

Let $A, B \in M^{m \times n}$ and a scalar α be given. Then we have

$$(i) \quad (A^T)^T = A$$

$$(ii) \quad (A + B)^T = A^T + B^T$$

$$(iii) \quad (\alpha A)^T = \alpha A^T$$

Square Matrices and Symmetric Matrices

An $m \times n$ matrix is said to be square if $m = n$, i.e. if the number of rows is the same as the number of columns.

A matrix A is said to be *symmetric* provided that $A^T = A$.

Notice that a symmetric matrix is necessarily square.

Symmetric matrices with real entries have a number of “magical” properties. We shall discuss some of these later on.

Products of Matrices

We could define the product of two matrices of the same size by simply multiply the corresponding entries with one another. This turns out not to be a very useful notion. One of the most important uses of $m \times n$ matrices is to associate them with linear mappings from V^n to V^m by multiplying the matrix by an element of V^n to produce an element of V^m . Matrix multiplication is designed to be consistent with this point of view.

In order for the product of two matrices to be defined we need the number of columns of the first matrix to match the numbers of rows of the second matrix. Let $A \in M^{m \times n}, B \in M^{n \times p}$ be given. We define $AB \in M^{m \times p}$ by

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj} \quad \text{for all } i = 1, 2, \dots, m, j = 1, 2, \dots, p.$$

Products of Matrices (Continued)

In particular, if $A \in M^{m \times n}$ and $x \in V^n$, then $Ax \in V^m$ is given by

$$(Ax)_i = \sum_{j=1}^n A_{ij}x_j \quad \text{for all } i = 1, 2, \dots, m.$$

Example: Let

$$A = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad x = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}. \quad \text{Let us}$$

calculate Bx and AB .

Example (Continued)

$$(Bx)_1 = 1(2) + 2(-1) + 3(3) = 9,$$

$$(Bx)_2 = 4(2) + 5(-1) + 6(3) = 21,$$

so we have

$$Bx = \begin{pmatrix} 9 \\ 21 \end{pmatrix} \in V^2.$$

The product AB will be a 2×3 matrix.

Example (Continued)

$$(AB)_{11} = 2(1) + (-1)(4) = -2, \quad (AB)_{12} = 2(2) + (-1)5 = -1,$$

$$(AB)_{13} = 2(3) + (-1)6 = 0, \quad (AB)_{21} = (-3)1 + 2(4) = 5,$$

$$(AB)_{22} = (-3)2 + 2(5) = 4, \quad (AB)_{23} = (-3)3 + 2(6) = 3.$$

$$AB = \begin{pmatrix} -2 & -1 & 0 \\ 5 & 4 & 3 \end{pmatrix}$$

Properties of Matrix Multiplication

One of the most important properties of matrix multiplication is really a “non-property”; more precisely it is the observation that matrix multiplication is not commutative. In general the fact that AB is defined does not even imply that BA is defined. Moreover, even if AB and BA both are defined, they may not be of the same size.

In the special case when $A, B \in M^{n \times n}$ then AB and BA are both defined and they have the same size, namely $n \times n$. However, it can easily happen that $AB \neq BA$.

Example:
$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$$

Associative Property of Matrix Multiplication

Matrix multiplication is associative. More precisely, let $A \in M^{m \times n}$, $B \in M^{n \times p}$, and $C \in M^{p \times k}$ be given. Then we have $AB \in M^{m \times p}$ and $BC \in M^{n \times k}$. Thus the products $(AB)C$ and $A(BC)$ are defined and both belong to $M^{m \times k}$. In fact we have

$$(AB)C = A(BC),$$

as can be verified by a straightforward, but tedious, calculation.

If α is a scalar then

$$\alpha(AB) = (\alpha A)B = A(\alpha B).$$

Distributive Properties of Matrix Multiplication

Matrix multiplication satisfies distributive properties so long as we are careful to keep products in the correct order. Let $E \in M^{m \times n}$, $F, G \in M^{n \times p}$, $H \in M^{p \times k}$ be given and let α be a scalar.

Then we have

$$E(F + G) = EF + EG,$$

$$(F + G)H = FH + GH,$$

$$\alpha(F + G) = (\alpha F) + (\alpha G).$$

Identity Matrix

The $m \times m$ identity matrix, denoted by I_m is defined by

$$(I_m)_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j, \end{cases}$$

i.e.

$$I_m = \begin{pmatrix} 1 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot & \cdot & 1 \end{pmatrix}.$$

Identity Matrix (Continued)

It is customary to write I in place of I_m if m is clear from context.

Observe that if $A \in M^{m \times n}$ then

$$AI_n = A = I_mA$$

Linear Transformations

Let $M \in A^{m \times n}$ be given. Then for each $x \in V^n$ we have $Ax \in V^m$.
The mapping

$$x \rightarrow Ax$$

takes V^n to V^m . (The range of this mapping is generally not all of V^m .)

It is extremely useful to identify matrices with mappings or transformations from V^n to V^m rather than just arrays of numbers. Such transformations are linear in the sense that

- ▶ $A(x + y) = Ax + Ay$ for all $x, y \in V^n$,
- ▶ $A(\alpha x) = \alpha Ax$ for all scalars α and all $x \in V^n$.

Matrix Multiplication and Composition of Functions

Let $A \in M^{m \times n}$ and $B \in M^{n \times p}$ be given and define $f : V^n \rightarrow V^m$ and $g : V^p \rightarrow V^n$ by

$$f(y) = Ay \text{ for all } y \in V^n, \quad g(x) = Bx \text{ for all } x \in V^p.$$

Then we have

$$f(g(x)) = A(Bx) = (AB)x \text{ for all } x \in V^p.$$

Covariance Matrices

Let X be a random vector with n components, i.e.

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \cdot \\ \cdot \\ \cdot \\ X_n \end{pmatrix}.$$

The covariance matrix $\mathbb{C}(X)$ of X is the $n \times n$ matrix defined by

$$(\mathbb{C}(X))_{ij} = \text{Cov}(X_i, X_j), \quad i, j = 1, 2, \dots, n.$$

Observe that $\mathbb{C}(X)$ is always symmetric.

Covariance Matrices (Cont.)

Let $A \in M^{m \times n}$ be given and define a new random vector Y by

$$Y = AX.$$

Then we have

$$\mathbb{C}(Y) = AC(X)A^T.$$

Subspaces

Definition: A set $W \subset V^k$ is called a *subspace* of V^k provided that $W \neq \emptyset$ and

$$c_1x + c_2y \in W \text{ for all } x, y \in W \text{ and all scalars } c_1, c_2.$$

(Every subspace must contain the zero vector.)

Let $A \in M^{m \times n}$ be given. There are two extremely important subspaces associated with A .

Definition: The *null space* $\mathcal{N}(A)$ of A is defined by

$$\mathcal{N}(A) = \{x \in V^n : Ax = 0\}.$$

Definition: The *range* $\mathcal{R}(A)$ of A is defined by

$$\mathcal{R}(A) = \{Ax : x \in V^n\}.$$

Subspaces (Continued)

The null space and range of a matrix A provide us with a lot of important information. In particular, let $A \in M^{m \times n}$ and $b \in V^m$ be given. Then:

- ▶ The system of equations $Ax = b$ has a solution $x \in V^n$ if and only if $b \in \mathcal{R}(A)$. (In particular, $Ax = b$ will be solvable for every $b \in V^m$ if and only if $\mathcal{R}(A) = V^m$.)
- ▶ If $\mathcal{N}(A) = \{0\}$ then the system of equations $Ax = b$ has at most one solution.

Linear Combinations and Spans

Consider a list $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ of vectors in V^k .

Definition: By a *linear combination* of $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ we mean an expression of the form

$$c_1 v^{(1)} + c_2 v^{(2)} + \dots + c_p v^{(p)}.$$

Definition: The set of all linear combinations of $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ is called the *span* of $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ and is denoted by $\text{span}(v^{(1)}, v^{(2)}, \dots, v^{(p)})$.

Remark: $\text{span}(v^{(1)}, v^{(2)}, \dots, v^{(p)})$ is a subspace of V^k .

Example: Let $k = 3$ and put

$$v^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

If we form a linear combination of $v^{(1)}, v^{(2)}$ with scalars c_1, c_2 we get

$$c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_2 \end{pmatrix}.$$

We see that

$$\text{span}(v^{(1)}, v^{(2)}) = \{v \in V^3 : v_2 = v_3\}.$$

It is not difficult to see that

$$\text{span}\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = V^3$$

Dimension of a Subspace

Let k be a positive integer and let W be a subspace of V^k .

Fact: There is a finite list of vectors $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ such that

$$\text{span}(v^{(1)}, v^{(2)}, \dots, v^{(p)}) = W.$$

Notice that if we append an additional vector from W to such a list, the span of the enlarged list will still be W .

Definition: The smallest positive integer p such that there is a list of vectors $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ having $\text{span}(v^{(1)}, v^{(2)}, \dots, v^{(p)}) = W$ is called the *dimension* of W and is denoted $\dim(W)$.

Dimension of a Subspace (Continued)

Fact: $\dim(V^k) = k$

Remark: We make the convention that $\dim(\{0\}) = 0$.

The Rank and Nullity Theorem

Let $A \in M^{m \times n}$ be given. Then

$$\dim(\mathcal{N}(A)) + \dim(\mathcal{R}(A)) = n.$$

Example: Let

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Then for all $x \in V^3$ we have

$$Ax = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + x_3 \\ x_2 + x_3 \\ x_2 + x_3 \end{pmatrix}.$$

Example Continued

Observe that

$$Ax = (x_1 + x_3) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + (x_2 + x_3) \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

We see that

$$\mathcal{R}(A) = \text{span}\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right).$$

We can conclude that $\dim(\mathcal{R}(A)) = 2$. It follows from the Rank and Nullity Theorem that $\dim(\mathcal{N}(A)) = 1$.

Example Continued

Let's find $\mathcal{N}(A)$ directly. We look for $x \in V^3$ such that

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

This is equivalent to the pair of equations

$$x_1 + x_3 = 0, \quad x_2 + x_3 = 0.$$

We see that $x \in \mathcal{N}(A)$ if and only if $x_1 = x_2 = -x_3$, i.e. if and only

$$x = x_3 \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

We conclude that

$$\mathcal{N}(A) = \text{span}\left(\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}\right).$$

Rank of a Matrix

Fact: If $A \in M^{m \times n}$ then

$$\dim(\mathcal{R}(A)) \leq m \text{ and } \dim(\mathcal{R}(A)) \leq n.$$

Definition: $\dim(\mathcal{R}(A))$ is called the *rank* of A . Notice that

$$\text{rank}(A) \leq \min\{m, n\}.$$

($\dim(\mathcal{N}(A))$ is sometimes called the *nullity* of A , although this terminology is not used very often, except in the name “rank and nullity theorem”.)

Definition: A matrix $A \in M^{m \times n}$ is said to have *full rank* provided that

$$\text{rank}(A) = \min\{m, n\}.$$

Column Space and Range

Before looking at systems of linear equations, we make a useful observation about the range of a matrix. Let $A \in M^{m \times n}$ be given. Then for every $x \in V^n$ we have

$$Ax = x_1 \begin{pmatrix} A_{11} \\ A_{21} \\ \vdots \\ A_{m1} \end{pmatrix} + x_2 \begin{pmatrix} A_{12} \\ A_{22} \\ \vdots \\ A_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} A_{1n} \\ A_{2n} \\ \vdots \\ A_{mn} \end{pmatrix}.$$

We see that the range of A is the span of the columns of A . For this reason the range of A is often called the *column space* of A .

Systems of Linear Equations

Consider the system of linear equations

$$\begin{array}{rcl} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n & = & b_1 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n & = & b_2 \\ \quad \quad \quad \cdot \cdot \cdot \cdot \cdot & = & \cdot \\ \quad \quad \quad \cdot \cdot \cdot \cdot \cdot & = & \cdot \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n & = & b_m \end{array}$$

This system can be written more compactly as

$$(L) \quad Ax = b.$$

Here $A \in M^{m \times n}$ and $b \in V^m$ are given and we seek a solution $x \in V^n$. It is useful to introduce the associated homogeneous system

$$(H) \quad Ax = 0.$$

The zero vector is always a solution of the homogeneous system (H). The set of all solutions of (H) is precisely the null space $\mathcal{N}(A)$.

Observe that (L) has at least one solution if and only if $b \in \mathcal{R}(A)$.

Principle of Superposition

Recall that $A \in M^{m \times n}$

- ▶ The set of all solutions of (H) is a subspace of V^n .
- ▶ If $x^{(1)}$ and $x^{(2)}$ are solutions of (L) then $x^{(1)} - x^{(2)}$ is a solution of (H)
- ▶ If x is a solution of (L) and $x^{(H)}$ is a solution of (H) then $x + x^{(H)}$ is a solution of (L).

Remark: If $\text{rank}(A) = m$ then (L) has at least one solution. If $\text{rank}(A) = n$ then $\dim(\mathcal{N}(A)) = 0$ (by the rank and nullity theorem) so that (H) has only the zero solution and this implies that (L) has at most one solution. If $\text{rank}(A) < n$ then $\dim(\mathcal{N}(A)) > 0$ so (H) has infinitely many solutions.

Count of Equations versus Unknowns

Many times students want to draw quick and easy conclusions about the nature of solutions by simply comparing the number of equations to the number of unknowns. Some care must be exercised here. There is one easy conclusion of this type that we observe here: If the number of unknowns strictly exceeds the number of equations then (H) has infinitely many solutions. This is because $\text{rank}(A) \leq m < n$ so the rank and nullity theorem ensures that $\dim(\mathcal{N}(A)) > 0$. One cannot draw any implications on the nature of solutions from the fact that the number of equations equals the number of unknowns or from the fact that the number of equations strictly exceeds the number of unknowns.

Linear Independence

Consider a list of vectors $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ in V^k . We say that this list of vector is *linearly dependent* if there exist scalars $c_1, c_2, \dots, c_p$ at least one of which is nonzero such that

$$c_1 v^{(1)} + c_2 v^{(2)} + \dots + c_p v^{(p)} = 0.$$

It is also common to say that the vectors are linearly dependent (instead of saying that the list is linearly dependent). Linearly dependent means that there is some redundancy in the list of vectors. More precisely, the list is linearly dependent if and only if at least one the vectors can be expressed as a linear combination of the others.

We say that the list is *linearly independent* if it is not linearly dependent.

Linear Independence (Continued)

Fact: The list of vectors $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ is linearly independent if and only if

$$\dim(\text{span}(v^{(1)}, v^{(2)}, \dots, v^{(p)})) = p.$$

Fact: If $p > k$ then the list $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ is linearly dependent.

It is instructive to relate linear independence of a list of vectors to the null space of the matrix having those vectors as columns. To this end define $A \in M^{p \times k}$ by

$$A_{ij} = v_i^{(j)}, \quad i = 1, 2, \dots, k, \quad j = 1, 2, \dots, p.$$

The list $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ of vectors is linearly independent if and only if $\mathcal{N}(A) = \{0\}$.

Example

Determine whether each list of vectors is linearly independent in $\mathbb{R}^3$.

(a) The vectors $u^{(1)} = \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$, $u^{(2)} = \begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix}$,
 $u^{(3)} = \begin{pmatrix} -1 \\ -1 \\ 7 \end{pmatrix}$, $u^{(4)} = \begin{pmatrix} 2 \\ 0 \\ -5 \end{pmatrix}$ are linearly dependent by inspection because the number of vectors (*four*) exceeds the dimension of the space (*three*).

(b) The vectors $v^{(1)} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$, $v^{(2)} = \begin{pmatrix} -5 \\ -4 \\ 2 \end{pmatrix}$ are linearly independent by inspection because neither one is a constant multiple of the other.

Example (Continued)

(c) In order to decide whether the vectors

$$w^{(1)} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad w^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}, \quad w^{(3)} = \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix}$$

are linearly independent, we will need to do some work. The condition

$$c_1 w^{(1)} + c_2 w^{(2)} + c_3 w^{(3)} = 0$$

is equivalent to the system of equations

$$(1) \quad c_1 - c_3 = 0$$

$$(2) \quad 2c_1 + c_2 - 3c_3 = 0$$

$$(3) \quad c_1 + 4c_2 + 2c_3 = 0.$$

Example (Continued)

Equation (1) tells us that $c_3 = c_1$. Substituting this into (2) we see that $c_2 = c_1$. Substituting $c_1 = c_2 = c_3$ into (3) we find that $7c_1 = 0$. Consequently $c_1 = c_2 = c_3 = 0$ and the vectors are linearly independent.

We will revisit part (c) of this example later on and approach it using determinants.

Basis for a Subspace

Let W be a subspace of V^n .

Definition: A list $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ of vectors in W is called a *basis* for W provided this list is linearly independent and $\text{span}(v^{(1)}, v^{(2)}, \dots, v^{(p)}) = W$.

Fact: Every subspace has at least one basis. All bases for a given subspace have the same number of vectors, namely the dimension of W .

Fact: Let $v^{(1)}, v^{(2)}, \dots, v^{(p)}$ be a basis for W . Then for each $x \in W$ there is exactly one list $c_1, c_2, \dots, c_p$ of scalars such that

$$x = c_1 v^{(1)} + c_2 v^{(2)} + \dots + c_p v^{(p)}.$$

Deep Properties of the Transpose

Let $A \in M^{m \times n}$ be given. Recall that the transpose $A^\top$ of A is the $n \times m$ matrix defined by

$$(A^\top)_{ij} = A_{ji} \quad \text{for all } i = 1, 2, \dots, n, j = 1, 2, \dots, m.$$

For example

$$\text{If } A = \begin{pmatrix} 1 & 2 & -3 \\ 5 & 0 & 6 \end{pmatrix} \quad \text{then } A^\top = \begin{pmatrix} 1 & 5 \\ 2 & 0 \\ -3 & 6 \end{pmatrix}.$$

Deep Properties of the Transpose (Continued)

Fact: Let $A \in M^{m \times n}$ and $B \in M^{n \times k}$ be given. Then

$$(AB)^{\top} = B^{\top} A^{\top}.$$

There is a pair of very important relationships involving null spaces, ranges, and transposes for matrices with real entries. (Similar relationships hold for matrices with complex entries, but they involve complex conjugates.)

Fact: Let A be an $m \times n$ matrix with real entries. Then

- ▶ $\mathcal{R}(A) = \{y \in V^m : y^{\top} x = 0 \text{ for all } x \in \mathcal{N}(A^{\top})\}$
- ▶ $\mathcal{N}(A) = \{x \in V^n : x^{\top} y = 0 \text{ for all } y \in \mathcal{R}(A^{\top})\}$

Symmetric Matrices

A matrix $A \in M^{m \times n}$ is said to be symmetric provided $A = A^T$.
(This requires that $m = n$.)

Covariance matrices are symmetric.

Hessian matrices (of smooth functions) are symmetric by equality of mixed partial derivatives.

Warning: $A, B \in M^{n \times n}$ and A, B both symmetric does not imply that AB is symmetric!

Diagonal Matrices

A matrix $D \in M^{n \times n}$ is said to be diagonal provided $D_{ij} = 0$ whenever $i \neq j$.

Every diagonal matrix is symmetric.

The identity matrix is diagonal.

Warning: $A \in M^{n \times n}$, $D \in M^{n \times n}$, and D diagonal does not imply that $AD = DA$, even if A is symmetric.

Inverses

An $n \times n$ matrix A is said to be invertible provided there exists $B \in M^{n \times n}$ such that

$$AB = BA = I.$$

Here I is the $n \times n$ identity matrix. For a given matrix A there can be at most one such B . We write $B = A^{-1}$.

Suppose that A is invertible and consider the linear system

$$(L) \quad Ax = b,$$

where $b \in V^n$ is given. Then we have

$$\begin{aligned} A^{-1}(Ax) &= A^{-1}b \\ (A^{-1}A)x &= A^{-1}b \\ Ix &= A^{-1}b \\ x &= A^{-1}b. \end{aligned}$$

Properties of Inverses

Let $A, B \in M^{n \times n}$ be given.

- ▶ If A is invertible then so is A^{-1} and $(A^{-1})^{-1} = A$.
- ▶ If A is invertible then so is A^T and $(A^T)^{-1} = (A^{-1})^T$.
- ▶ If A is invertible and c is a nonzero scalar, then cA is invertible and $(cA)^{-1} = c^{-1}A^{-1}$.
- ▶ If A and B both are invertible then so is AB and $(AB)^{-1} = B^{-1}A^{-1}$.
- ▶ A is invertible if and only if $\text{rank}(A) = n$.

One-Sided Inverses are Actually Two-sided Inverses

Let $A, B \in M^{n \times n}$ be given and assume that

$$AB = I.$$

Then it is also true that

$$BA = I.$$

Consequently:

- ▶ A is invertible and $A^{-1} = B$.
- ▶ B is invertible and $B^{-1} = A$.

Elementary Row Operations

There are 3 basic elementary row operations that can be performed on matrices without changing the rank of the matrix:

- ▶ Interchange two rows
- ▶ Add a scalar multiple of one row to a different row
- ▶ Multiple a row by a nonzero scalar

Using these row operations provides a systematic way of determining the rank of a matrix, finding the inverse of an invertible matrix, and solving systems of linear equations.

Finding an Inverse via Row Operations

We illustrate the use of row operations by finding the inverse of

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 3 & -1 & -2 \\ -8 & 4 & 5 \end{pmatrix}.$$

For this purpose, we construct an *augmented matrix* having A on the left and the 3×3 identity I to the right of the “separating bar”:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 3 & -1 & -2 & 0 & 1 & 0 \\ -8 & 4 & 5 & 0 & 0 & 1 \end{array} \right).$$

The idea is to perform elementary row operations to produce the identity matrix to the left of the bar.

$$R_2 - 3R_1 \rightarrow R_2 \quad \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -3 & 1 & 0 \\ -8 & 4 & 5 & 0 & 0 & 1 \end{array} \right)$$

$$R_3 + 8R_1 \rightarrow R_3 \quad \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -3 & 1 & 0 \\ 0 & 4 & -3 & 8 & 0 & 1 \end{array} \right)$$

$$R_3 + 4R_2 \rightarrow R_3 \quad \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -3 & 1 & 0 \\ 0 & 0 & 1 & -4 & 4 & 1 \end{array} \right)$$

$$R_1 + R_3 \rightarrow R_1 \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 4 & 1 \\ 0 & -1 & 1 & -3 & 1 & 0 \\ 0 & 0 & 1 & -4 & 4 & 1 \end{array} \right)$$

$$R_2 - R_3 \rightarrow R_2 \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 4 & 1 \\ 0 & -1 & 0 & 1 & -3 & -1 \\ 0 & 0 & 1 & -4 & 4 & 1 \end{array} \right)$$

$$-R_2 \rightarrow R_2 \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 4 & 1 \\ 0 & 1 & 0 & -1 & 3 & 1 \\ 0 & 0 & 1 & -4 & 4 & 1 \end{array} \right)$$

Now that we have the identity to the left of the dividing bar, the matrix to the right of the dividing bar will be A^{-1} . We conclude that

$$A^{-1} = \begin{pmatrix} -3 & 4 & 1 \\ -1 & 3 & 1 \\ -4 & 4 & 1 \end{pmatrix}.$$

It is straightforward to check that

$$AA^{-1} = A^{-1}A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Row Echelon Form

In order to use the elementary row operations to determine the rank of a matrix, it is helpful to introduce the idea of row echelon form of a matrix. When we speak of a row being 0, we mean that all entries are 0; when we speak of a nonzero row, we mean that at least one entry is not 0.

Definition: A matrix is said to be in *row echelon form* provided all rows of zeros are at the bottom and the first nonzero entry of each nonzero row is strictly to the left of the first nonzero entries of the nonzero rows below the row in question.

Definition: A matrix is said to be in *reduced row echelon form* if it is in row echelon form and the following two additional properties hold: (i) the first nonzero entry in each row is 1; and (ii) the first nonzero entry in each row has no other nonzero entries in the same column.

Using Row Operations to Determine Rank

The 3 elementary row operations do not change the rank of a matrix. In order to determine the rank of $A \in M^{m \times n}$, we can start with A and keep performing row operations until the matrix is in row echelon form. The number of nonzero rows will be equal to the rank of the original matrix.

Example: Let's find the rank of

$$\begin{pmatrix} 1 & 2 & 3 & -1 \\ 4 & 2 & -3 & 5 \\ 6 & 6 & 3 & 3 \end{pmatrix}$$

by reducing it to row echelon form.

$$R_2 - 4R_1 \rightarrow R_2 \quad \begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & -6 & -15 & 9 \\ 6 & 6 & 3 & 3 \end{pmatrix}$$

$$R_3 - 6R_1 \rightarrow R_3 \quad \begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & -6 & -15 & 9 \\ 0 & -6 & -15 & 9 \end{pmatrix}$$

$$R_3 - R_2 \rightarrow R_3 \quad \begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & -6 & -15 & 9 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since there are two nonzero rows in the row echelon form, the rank of the original matrix is 2.

Some Properties of Rank

In addition to the rank and nullity theorem, there are several important results about rank that you should be aware of. Let $A, B \in M^{m \times n}$ and $C \in M^{n \times k}$ be given. Notice that AA^T and $A^T A$ both are defined. We have

- ▶ $\text{rank}(A) \leq \min\{m, n\}$
- ▶ $\text{rank}(AC) \leq \min\{\text{rank}(A), \text{rank}(C)\}$
- ▶ $\text{rank}(A) + \text{rank}(C) \leq \text{rank}(AC) + n$
- ▶ $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$
- ▶ $\text{rank}(A) = \text{rank}(A^T)$
- ▶ If A has real entries then
$$\text{rank}(A^T A) = \text{rank}(AA^T) = \text{rank}(A) = \text{rank}(A^T).$$

Determinants

The *determinant function* assigns to each square matrix $A \in M^{n \times n}$ a scalar $\det(A)$, called the determinant of A . If the entries of A are real, then $|\det(A)|$ is the n -dimensional volume of the parallelotope determined by the column vectors of A . One very important feature of the determinant is that $\det(A) = 0$ if and only if the columns of A are linearly dependent.

We start by looking at the 2×2 case. For

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

we have

$$\det(A) = A_{11}A_{22} - A_{12}A_{21}.$$

Determinant of a 3×3 Matrix

Let $A \in M^{3 \times 3}$ be given. For each $i, j = 1, 2, 3$ let us write $\mathcal{A}^{(ij)}$ for the 2×2 matrix obtained from A by deleting row i and column j from A .

Then we have

$$\det(A) = A_{11}\det(\mathcal{A}^{11}) - A_{12}\det(\mathcal{A}^{12}) + A_{13}\det(\mathcal{A}^{13}).$$

Example: Let

$$A = \begin{pmatrix} 1 & 3 & -1 \\ 3 & -1 & -2 \\ -8 & 4 & 5 \end{pmatrix}.$$

Example Continued

Observe that

$$\mathcal{A}^{(11)} = \begin{pmatrix} -1 & -2 \\ 4 & 5 \end{pmatrix}, \quad \mathcal{A}^{(12)} = \begin{pmatrix} 3 & -2 \\ -8 & 5 \end{pmatrix}, \quad \mathcal{A}^{(13)} = \begin{pmatrix} 3 & -1 \\ -8 & 4 \end{pmatrix}.$$

We have

$$\det(A) = 1 \cdot \det(\mathcal{A}^{(11)}) - 3 \cdot \det(\mathcal{A}^{(12)}) + (-1) \cdot \det(\mathcal{A}^{(13)}).$$

It follows that

$$\det(A) = (-5 + 8) - 3(15 - 16) - (12 - 8) = 2.$$

Determinant of a 3×3 Diagonal Matrix

Let

$$A = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix},$$

where $\lambda_1, \lambda_2, \lambda_3$ are given real numbers.

A simple computation shows that

$$\det(A) = \lambda_1 \lambda_2 \lambda_3.$$

The parallelepiped determined by the column vectors of A is a “rectangular box” with sides $|\lambda_1|, |\lambda_2|, |\lambda_3|$.

Determinants and Linear Independence

Since the absolute value of the $\det(A)$ is the volume of the parallelepiped determined by the column vectors of A , we have the following nice result concerning linear independence: The column vectors of A are linearly independent if and only if

$$(4) \quad \det(A) \neq 0.$$

(It is also that the rows of A are linearly independent if and only if (4) holds, because the determinant of a matrix is the same as the determinant of the transpose.)

Example

Earlier, we saw that the vectors

$$w^{(1)} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad w^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}, \quad w^{(3)} = \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix}$$

are linearly independent by solving a system of 3 linear equations. Let's establish linear independence by using the determinant. Let

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -3 \\ 1 & 4 & 2 \end{pmatrix}.$$

We see that

$$\det(A) = 1(2 + 12) - 0 - 1(8 - 1) = 7.$$

We conclude that the vectors are linearly independent.